

Plant

KodaPure · CAT # KBTNA-P-26

IMPORTANT

NOTE BEFORE START

- 1. Mix all buffers and suspensions thoroughly immediately before use. Reagents such as MB, PW, and LBIII may contain settled or undissolved material; resuspend completely to ensure consistent pipetting and reliable performance.
- 2. For samples with high protein content, add Proteinase K during the 56 °C incubation step, where applicable after LBII has been added. Use Proteinase K and follow the validated amount for your sample type.
- 3. To isolate DNA only, treat the sample with RNase A as described in the Notes section.
- 4. If the LBII bottle shows a substantial reduction in volume before it has been opened, do not use the reagent. Report the issue to KreatBio and request a replacement.

PROTOCOL SUMMARY

SAMPLES	SAMPLE INPUT	HOMOGENIZATION	INHIBITOR REMOVAL
Leaf, seed, root, stem, and other plant tissue	50-100 mg	In LBIII at 65 °C (see below)	Tube A, 3 min
ELUATE STORAGE			
4 °C short / -20 °C long			

REAGENTS SUPPLIED

● 1 X 12 mL LBI Lysis Buffer I	● 1 X 37 mL LBIII Lysis Buffer III
● 50 X HB045 Homogenizing Beads 0.5 mm & 4 mm	● 50 X Tube A Inhibitor-removal tube
● 1 X 22 mL BB Binding Buffer	● 1 X 1.1 mL MB Magnetic Beads
● 1 X 37 mL WBIII Wash Buffer III	● 1 X 72 mL WBI Wash Buffer II
● 1 X 5 mL EB Elution Buffer	

ADDITIONAL MATERIALS AND APPARATUS REQUIRED

- | | |
|--|---|
| <ul style="list-style-type: none">• Centrifuge machine suitable for 1.5 mL microcentrifuge tubes | <ul style="list-style-type: none">• 1.5 mL microcentrifuge tubes |
| <ul style="list-style-type: none">• Vortex machine or homogenizer | <ul style="list-style-type: none">• Pipettes and compatible pipette tips |
| <ul style="list-style-type: none">• Magnetic rack | <ul style="list-style-type: none">• Heat block or incubator capable of 65 °C |
| <ul style="list-style-type: none">• Tube rack | <ul style="list-style-type: none">• Optional, depending on homogenization method: liquid nitrogen with chilled mortar and pestle, or homogenizing stick |

METHOD

1. Select one homogenization method and process the plant sample in LBIII before continuing.

Cryogenic grinding

For high-quality, long-strand DNA: flash-freeze 50–100 mg sample in liquid nitrogen and grind to a fine powder using a chilled mortar and pestle. Mix LBIII thoroughly before use. In a 1.5 mL tube, add 700 μ L pre-warmed LBIII and the grinded sample. Mix well and incubate at 65 °C for 40 min.

High-speed shaking with homogenizing beads (optional: use KreatBio HB045)

For high-throughput processing: pre-warm LBIII and mix thoroughly. Add 500 μ L LBIII and one to two 3–5 mm homogenizing beads to a 1.5 mL tube. Add 50 mg sample cut into small pieces, then grind with a homogenizing stick until a slurry forms. If no stick is available, cut the tissue as finely as possible before adding it. Add another 200 μ L LBIII and mix. Incubate at 65 °C for 15 min, vortex at 3000–4000 rpm for 4 min until uniform, then incubate again at 65 °C for 40 min.

2. Centrifuge the tube at max speed for 5–10 min. Then, transfer 300 μ L of clear supernatant to a new 1.5 mL tube. (Avoid taking the dirty pellet.) Open one Tube A and transfer all Tube A contents to the transferred supernatant. Mix vigorously by hand and leave for 3 min. Centrifuge for another 5 min.
3. Transfer 200 μ L of the treated supernatant into a new tube. Add 200 μ L LBI, close the tube lid and mix well. Leave it for 5 min.
4. Bring MB to room temperature and mix it well before use. Add 400 μ L BB and 20 μ L MB to the sample tube. Mix gently and leave for 10 min.
5. Place the tube on the magnetic rack until the supernatant is clear (~2 min with KreatBio rack). Carefully decant or pipette out the clear supernatant without disturbing the black beads. Leave a small amount if necessary to reduce bead loss, then remove the tube from the rack.
6. Add 700 μ L WBIII to the tube and mix well. Place it back on the magnetic rack for 1 min or until supernatant is clear. Discard the supernatant while still on the magnetic rack. Once done, remove tube from the rack.
7. Add 700 μ L WBII to the tube and mix well. Place it back on the magnetic rack for 1 min or until supernatant is clear. Discard the supernatant, remove the tube from the rack, and repeat this step (Step 7). Tip: A quick spin, then back on the rack, removes excess WBII if there is. Remove the tube from the rack.
8. Open the tube lid and leave it for 5 min to air-dry.
9. Add 50–80 μ L EB to the black beads in the tube. Mix thoroughly and leave for 10 min. If some beads remain stuck to the tube wall, mix as well as possible and proceed.
10. The DNA/RNA is contained in the eluate. Place the tube on the magnetic rack for approximately 1 min, then transfer only the clear supernatant to a new tube leaving the black beads.



NOTES

- LBIII with HB045 beads is used to break tough plant cell walls. Pre-warm LBIII and mix it thoroughly before use.
- Choose cryogenic grinding when long-strand DNA is preferred. Choose high-speed shaking with HB045 when processing many samples or when a faster mechanical workflow is preferred.

TROUBLESHOOTING

EVENT	LIKELY CAUSE	SUGGESTED FIX
Low DNA yield	Too much sample, or poor homogenization	Use less starting material and homogenise fully.
RNA in a DNA-only eluate	RNase A not used	Add RNase A and incubate as noted.
Bead carryover	Beads disturbed during transfer	Leave on the magnet until clear, then transfer carefully.
Brown eluate	Soil, fat, or debris carried over	Repeat extraction and transfer only clear lysate, avoid the pellet and fat layer. Centrifuge if it helps with separation. If this is for PCR amplification, dilute eluate to 1:10 to 1:50.
Poor PCR result	Inhibitor carryover or too much template	Dilute the eluate 1:10 to 1:50 and avoid debris.
Soil inhibition	Humic substances carried over	Use Tube A for inhibitor removal, or dilute the eluate to 1:10 to 1:50 and rerun PCR amplification.

SAFETY

SAFETY & HANDLING

- Research use only. Wear gloves, a lab coat, and eye protection.
- BB, WBII, and WBIII are flammable — keep away from heat and flame.
- LBI and WBIII contain guanidine (GuHCl) — do not mix with bleach.
- Dispose of waste according to local rules.
- TNA means total nucleic acid — these kits recover both DNA and RNA.